TABLE II
COPNAV OBSERVABILITY ANALYSIS RESULTS

Case	Observable?	Observable States	Time Step l
1	no	none	5
2	no	$m = 1$: none $m \geq 2$: x_r, y_r, $\dot{x}_r$, $\dot{y}_r$	5 4
3	no	δt_r, $\dot{\delta t}_r$	5
4	yes	all	4
5	no	$\dot{x}_{r_i}$, $\dot{y}_{r_i}$, x_s, y_s, $i = 1, \ldots, n$	3
6	no	$\dot{x}_{r_i}$, $\dot{y}_{r_i}$, $i = 1, \ldots, n$	2
7	yes	all	2
8	yes	all	4

Theorem IV.7: A COpNav environment with one partially known receiver and one fully known SOP is observable $\forall\, l \geq 2$.

Proof: The state vector for this case is given by $\boldsymbol{x} = [\boldsymbol{x}_r^\mathsf{T},\ \boldsymbol{x}_s^\mathsf{T}]^\mathsf{T}$. Partial knowledge of the receiver and full knowledge of the SOP is equivalent to having an observation Jacobian matrix of the form

$$\mathbf{H}(t_k) = \begin{bmatrix} \mathbf{h}_{b,r,s}^\mathsf{T}(t_k) & \mathbf{h}_{c,r,s}^\mathsf{T}(t_k) \\ [\mathbf{I}_{2\times 2}\ \mathbf{0}_{2\times 4}] & \mathbf{0} \\ \mathbf{0} & \mathbf{I}_{4\times 4} \end{bmatrix}. \tag{25}$$

Invoking (15)–(20), it can be seen that $\text{rank}[\boldsymbol{\mathcal{O}}_L(t_0, t_1)] = 7$ since all the rows are linearly independent. Adding more time segments yields $\text{rank}[\boldsymbol{\mathcal{O}}_L(t_0, t_1)] = 10$ (full rank) $\forall l \geq 2$ since the first ten rows are linearly independent, whereas rows $4 + 7(l-1)$, $5 + 7(l-1)$, and $7 + 7(l-1)$ for $l = 1$ are identical to the corresponding rows for $l = 2, 3, \ldots$, and rows $\boldsymbol{\mathcal{O}}_{6+7(l-1)}^\mathsf{T}$ are linearly dependent, such that $\boldsymbol{\mathcal{O}}_{6+7(l-1)}^\mathsf{T} = \boldsymbol{\mathcal{O}}_6^\mathsf{T} + T(l-1)\boldsymbol{\mathcal{O}}_7^\mathsf{T}$. ■

Theorem IV.8: A COpNav environment with one fully known receiver and one unknown SOP is observable $\forall\, l \geq 4$.

Proof: The state vector for this case is given by $\boldsymbol{x} = [\boldsymbol{x}_r^\mathsf{T},\ \boldsymbol{x}_s^\mathsf{T}]^\mathsf{T}$. Full knowledge of the receiver is equivalent to having an observation Jacobian matrix of the form

$$\mathbf{H}(t_k) = \begin{bmatrix} \mathbf{h}_{b,r,s}^\mathsf{T}(t_k) & \mathbf{h}_{c,r,s}^\mathsf{T}(t_k) \\ \mathbf{I}_{6\times 6} & \mathbf{0} \end{bmatrix}. \tag{26}$$

Invoking (15)–(20), it can be seen that the observability matrix $\boldsymbol{\mathcal{O}}_L(t_0, t_l)$ has a rank of 7 at the first time segment since all the rows are linearly independent. The rank increments by one as each additional segment is appended up to $l = 4$. Adding a fourth time segment results in an observability matrix whose rank is 10 (full rank). This can be shown by noting that the first seven rows are linearly independent along with rows $8 + 7(l-2)$ for $l = 2, 3, 4$. Moreover, rows $i + 7(l-1)$ for $i = 4, 5, 7$ and $l = 1, 2, \ldots$, are identical, respectively. Finally, $\boldsymbol{\mathcal{O}}_{9+7(l-2)}^\mathsf{T} = \boldsymbol{\mathcal{O}}_2^\mathsf{T} + T(l-1)\boldsymbol{\mathcal{O}}_4^\mathsf{T}$, $\boldsymbol{\mathcal{O}}_{10+7(l-2)}^\mathsf{T} = \boldsymbol{\mathcal{O}}_3^\mathsf{T} + T(l-1)\boldsymbol{\mathcal{O}}_5^\mathsf{T}$, and $\boldsymbol{\mathcal{O}}_{13+7(l-2)}^\mathsf{T} = \boldsymbol{\mathcal{O}}_6^\mathsf{T} + T(l-1)\boldsymbol{\mathcal{O}}_7^\mathsf{T}$ for $l = 2, 3, \ldots$. ■

The results concluded from Theorems IV.1–IV.8 are summarized in Table II, where observable states refer to those in an orthogonal complement to the unobservable subspace, and time step l refers to the time step at which the observability matrix rank reaches a steady-state value. It is worth noting that the observability results for the scenarios considered in Table I constitute the minimal set of observability requirements in the sense that, by knowing the results for these scenarios, one can predict the observability of an arbitrary scenario with n receivers and m SOPs and any type of prior knowledge (none, partial, or full) for the receivers and SOPs.

V. SIMULATION RESULTS

Here, simulation results are presented, which were achieved for the three observable cases in Table I: Cases 4, 7, and 8. For purposes of numerical stability, the clock error states were defined to be $c\delta t$ and $c\dot{\delta t}$. All simulations assumed the receiver's process noise spectral density to be $\tilde{q}_x = \tilde{q}_y = 0.1\ \text{m}^2/\text{s}^4$, whereas the sampling period was set to $T = 10$ ms. The receiver's clock was assumed to be a temperature-compensated crystal oscillator with $h_0 = 2 \times 10^{-19}$ and $h_{-2} = 2 \times 10^{-20}$, whereas the SOPs' clocks were assumed to be oven-controlled crystal oscillators with $h_0 = 8 \times 10^{-20}$ and $h_{-2} = 4 \times 10^{-23}$. The observation noise spectral density was set to $r = 100\ \text{m}^2$.

A simulator was developed to generate the "truth" data for each COpNav environment studied. Noisy pseudorange observations were processed by an EKF to estimate the states of interest. The observability is quantified in terms of the estimation error $\tilde{\boldsymbol{x}} \triangleq \boldsymbol{x} - \hat{\boldsymbol{x}}$ and the corresponding estimation error covariance $\mathbf{P} \triangleq \mathbb{E}[\tilde{\boldsymbol{x}}\,\tilde{\boldsymbol{x}}^\mathsf{T}]$, where $\hat{\boldsymbol{x}}$ is the EKF state estimate. Results for a single-run EKF and rigorous MC analysis are presented. The MC analysis is based on an N-run average of the normalized estimation error squared (NEES) [31]. The ith-run NEES is defined as $\epsilon_i(t_k) \triangleq \tilde{\boldsymbol{x}}_i^\mathsf{T}(t_k|t_k)\mathbf{P}_i^{-1}(t_k|t_k)\tilde{\boldsymbol{x}}_i(t_k|t_k)$, whereas the average NEES is defined as $\bar{\epsilon}(t_k) = 1/N \sum_{i=1}^N \epsilon_i(t_k)$. For the single-run EKF, an observable system should yield converging estimation error covariances, and the estimation errors should remain bounded. For the N-run EKF, an observable system and a consistent EKF should yield a statistic $N\bar{\epsilon}(t_k)$ that is approximately chi-squared distributed with Nn degrees of freedom, specifically $N\bar{\epsilon}(k) \sim \chi^2_{Nn}$, where n is the state estimate dimension. An unobservable system should yield an estimation error covariance that never improves with more observations. Thus, the MC analysis boils down to a hypothesis test on $\bar{\epsilon}(t_k)$ with an acceptance region $[r_1, r_2]$ defined such that $\Pr\{\bar{\epsilon}(k) \in [r_1, r_2] | H_0\} = 1 - \alpha$, where H_0 is the null hypothesis, and α is the size of the test (probability of false alarm).

In the following simulations, the system true initial state $\boldsymbol{x}(t_0|t_{-1})$ was fixed, whereas the EKF initial state estimate $\hat{\boldsymbol{x}}(t_0)$ was generated according to $\hat{\boldsymbol{x}}(t_0|t_{-1}) \sim \mathcal{N}[\boldsymbol{x}(t_0), \mathbf{P}(t_0|t_{-1})]$, where $\mathbf{P}(t_0|t_{-1})$ is the EKF initial estimation error covariance. All the simulations assumed a receiver whose initial state is $\boldsymbol{x}_r(t_0) = [0, 0, 0, 25, 10, 1]^\mathsf{T}$ and SOPs with initial states $\boldsymbol{x}_{s_1}(t_0) = [50, 100, 1, 0.1]^\mathsf{T}$ and $\boldsymbol{x}_{s_2}(t_0) = [-50, 100, 1, 0.1]^\mathsf{T}$.

The simulations for Case 4 considered an environment with an unknown receiver and two SOPs: one fully known and one partially known. The initial estimation error covariance matrices of the receiver and the second SOP were chosen to be $\mathbf{P}_r(t_0|t_{-1}) = (1 \times 10^3)\text{diag}[2, 2, 1, 1, 30, 0.3]$ and $\mathbf{P}_{s_2}(t_0|t_{-1}) = (1 \times 10^3)\text{diag}[30, 0.3]$, respectively.

The simulations for Case 7 considered an environment with a partially known receiver and two SOPs: one fully known

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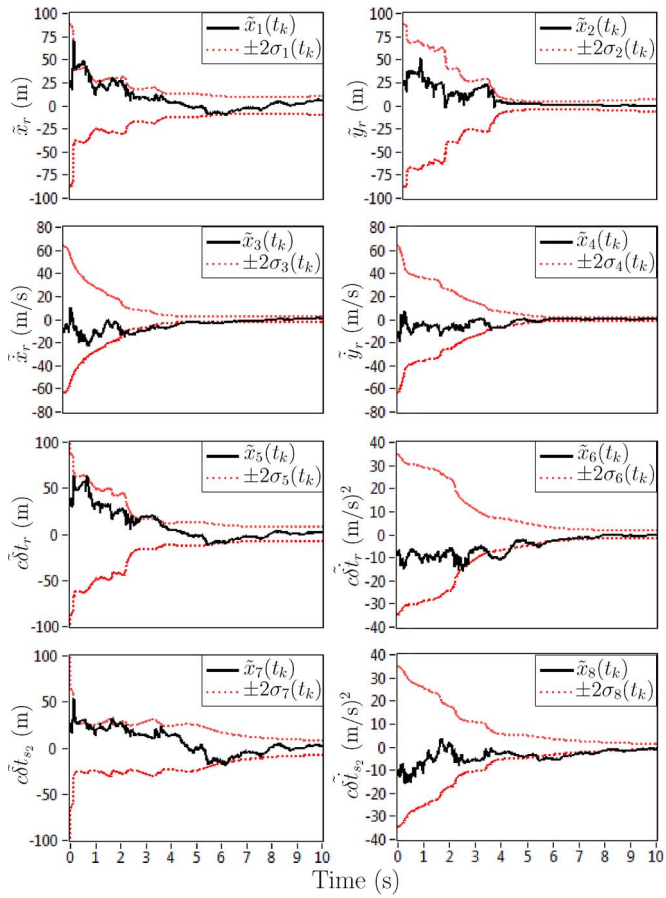

Fig. 1. Estimation error trajectories and $\pm 2\sigma$ bounds for Case 4 in Table I.

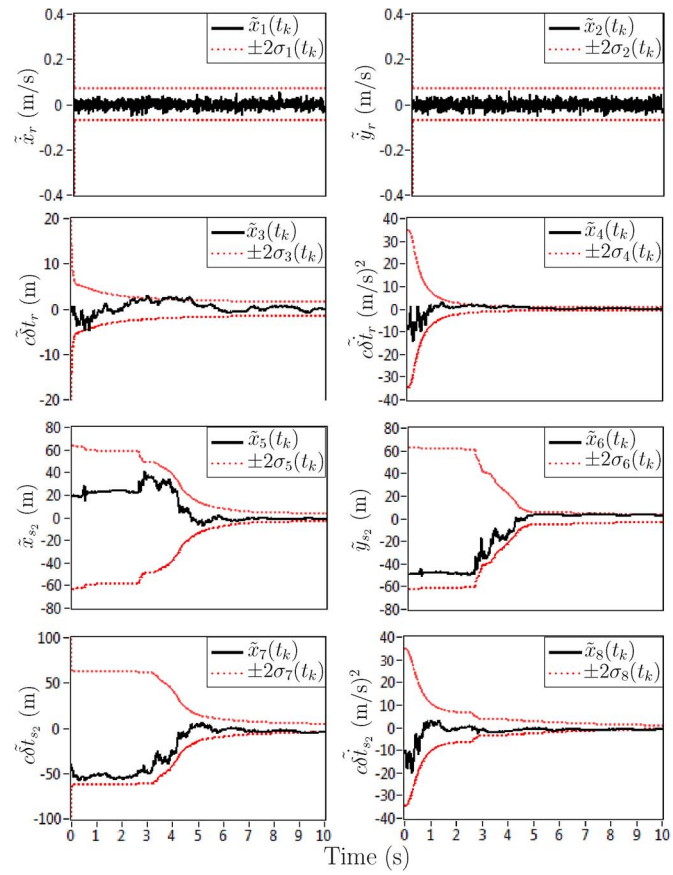

Fig. 2. Estimation error trajectories and $\pm 2\sigma$ bounds for Case 7 in Table I.

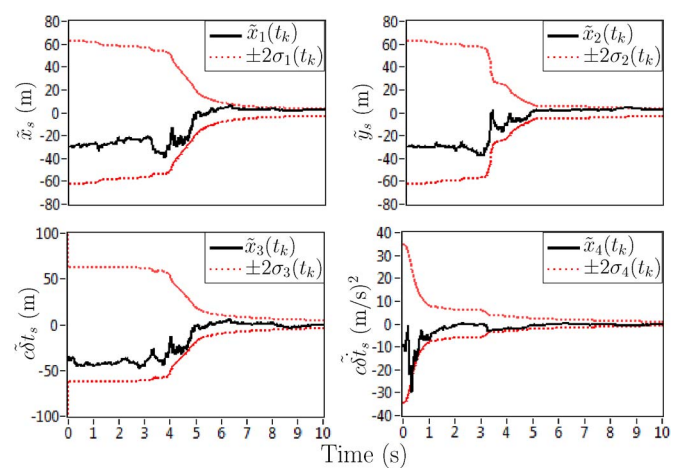

Fig. 3. Estimation error trajectories and $\pm 2\sigma$ bounds for Case 8 in Table I.

and one unknown. The initial estimation error covariance matrices of the receiver and the second SOP were chosen to be $\mathbf{P}_r(t_0|t_{-1}) = (1 \times 10^3)\mathrm{diag}[1, 1, 30, 0.3]$ and $\mathbf{P}_{s_2}(t_0|t_{-1}) = (1 \times 10^3)\mathrm{diag}[1, 1, 30, 0.3]$, respectively.

The simulations for Case 8 considered an environment with a fully known receiver and one unknown SOP. The initial estimation error covariance matrix of the SOP was chosen to be $\mathbf{P}_{s_1}(t_0|t_{-1}) = (1 \times 10^3)\mathrm{diag}[1, 1, 30, 0.3]$.

Figs. 1–3 show the estimation error trajectories $\tilde{x}_i(t_k|t_k)$ for a single-run EKF along with the $\pm 2\sigma_i(t_k|t_k)$ estimation error variance bounds for Cases 4, 7, and 8, respectively. Note that the estimation error variances converge and that the estimation errors remain bounded as would be expected for an observable system.

Figs. 4–6 show the resulting NEES trajectories $\bar{\epsilon}(t_k)$ for $\alpha = 0.01$ along with r_1 and r_2 for Cases 4, 7, and 8, respectively. Note that the $\bar{\epsilon}(t_k)$ values reside within the 99% probability region, which is consistent with a well-behaved estimator operating on an observable system.

VI. Experimental Results

A field experimental demonstration was conducted to illustrate one of the observable cases in Table I, namely, Case 8. The objective was to demonstrate that a COpNav receiver with velocity random-walk dynamics and knowledge of its initial state can estimate the states of an unknown SOP in its environment. To this end, two antennas were mounted on a vehicle to acquire and track: 1) multiple GPS signals; and 2) a signal from a nearby cellular phone tower whose signal was modulated through code-division multiple access (CDMA). The GPS and cellular signals were simultaneously downmixed and synchronously sampled via two National Instruments vector RF signal analyzers. These front ends fed their data to a Generalized Radionavigation Interfusion Device (GRID) software receiver [42], which simultaneously tracked all GPS L1 C/A signals in view and the signal from the cellular tower with unknown states, producing pseudorange observables for

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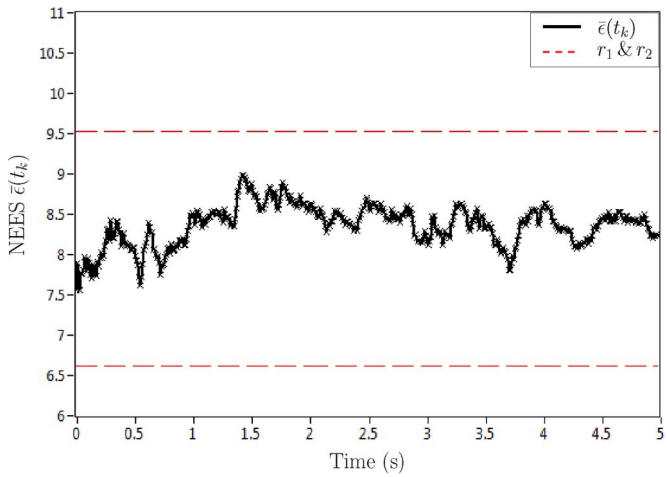

Fig. 4. NEES and r_1 and r_2 bounds for Case 4 in Table I with 50 MC runs.

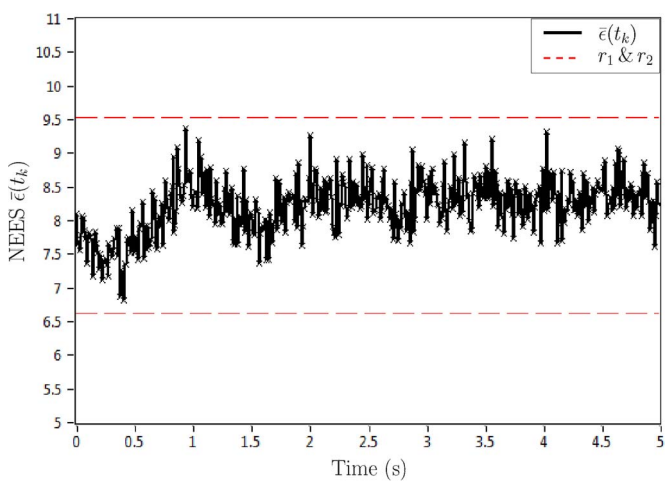

Fig. 5. NEES and r_1 and r_2 bounds for Case 7 in Table I with 50 MC runs.

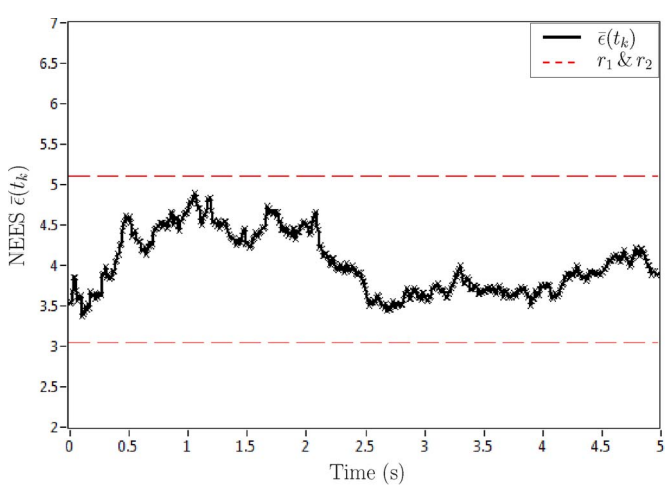

Fig. 6. NEES and r_1 and r_2 bounds for Case 8 in Table I with 50 MC runs.

all tracked signals. The observables were fed into a MATLAB-based EKF, which estimated the states of the unknown CDMA cellular tower. Fig. 7 shows the hardware setup of the conducted experiment.

Since the states of the GPS satellite vehicles (SVs) were known, and since the receiver was tracking more than four GPS SVs throughout the experiment, the receiver's initial state

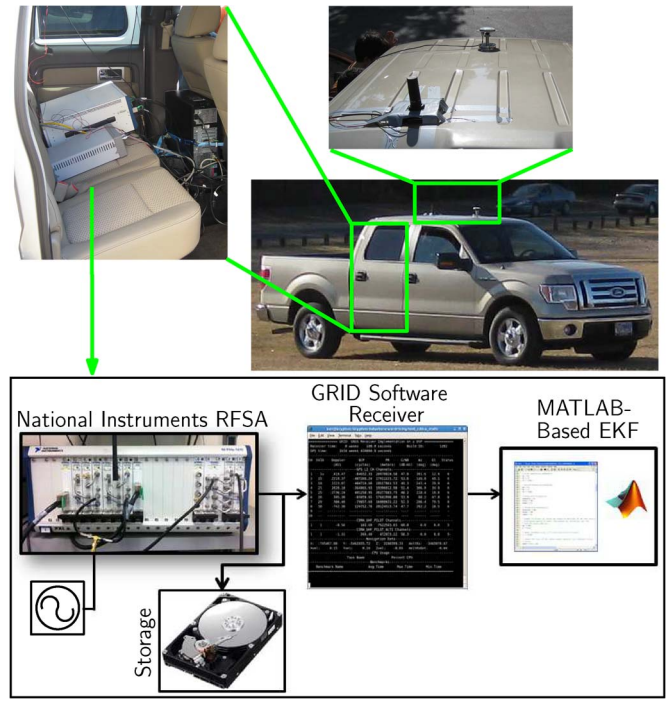

Fig. 7. Experiment hardware setup.

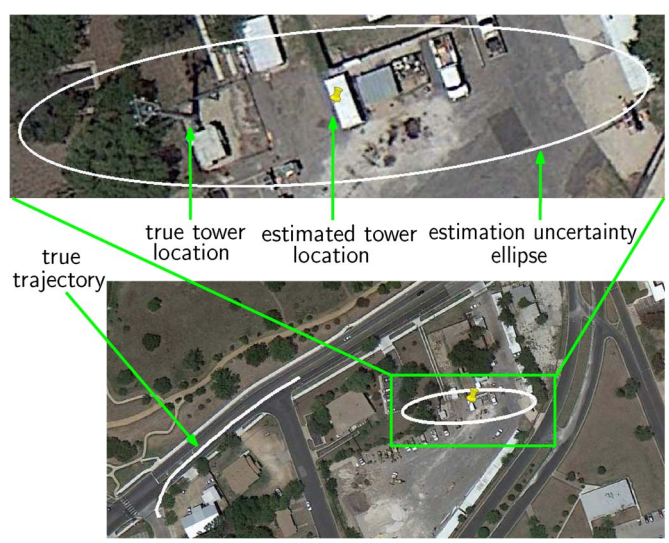

Fig. 8. Vehicle traversed trajectory during the collection of the GPS and cellular CDMA observations, true location of cellular CDMA tower, and estimated CDMA tower location and associated estimation error ellipse.

$\boldsymbol{x}_r(t_0)$ was fully known. The cellular tower state vector consisted of its planar position states, clock bias, and clock drift, as defined in (7). The EKF initial state estimate $\hat{\boldsymbol{x}}(t_0|t_{-1})$ was generated according to $\hat{\boldsymbol{x}}(t_0|t_{-1}) \sim \mathcal{N}[\boldsymbol{x}(t_0), \mathbf{P}(t_0|t_{-1})]$, where $\boldsymbol{x}(t_0) \triangleq [\boldsymbol{r}_s^{\mathsf{T}}(t_0), c\delta t_s(t_0), 0]^{\mathsf{T}}$, where $\boldsymbol{r}_s^{\mathsf{T}} \triangleq [x_s(t_0), y_s(t_0)]$ is the projection of the true cellular tower location from the Earth-Centered Earth-Fixed (ECEF) coordinate frame system to a planar system, $c\delta t_s(t_0) = \|\boldsymbol{r}_r(t_0) - \hat{\boldsymbol{r}}_s(t_0|t_{-1})\|_2 - \rho(t_0) + c\delta t_r(t_0)$, $\boldsymbol{r}_r^{\mathsf{T}}(t_0) \triangleq [x_r(t_0), y_r(t_0)]$ is the planar projection of the receiver's initial location from ECEF, and $\mathbf{P}(t_0|t_{-1}) = \mathrm{diag}[1 \times 10^4, 1 \times 10^4, 3 \times 10^4, 3 \times 10^2]$ is the EKF initial estimation error covariance matrix. Fig. 8 shows the receiver traversed trajectory during the collection of the pseudorange

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